FlexTome a 3D-printed, flexure-pivot rocking microtome with a differential-screw feed

Design study, revision 2. Companion to the full report (*flextome_paper.pdf*). All performance numbers are model predictions; the machine has not yet been built.

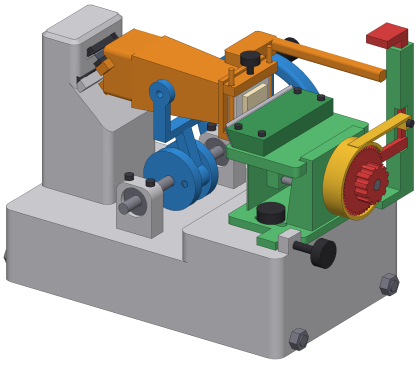
The problem. Diagnostic histology needs 3–5 µm paraffin sections. A new manual rotary microtome costs US$9 000–22 700; the thinnest published open-source sectioning device cuts at ≥50 µm. Much of sub-Saharan Africa has about one pathologist per million people, and the microtome is one of the few items in the diagnostic chain with no low-cost substitute.

The idea. Put the micrometre-scale motion entirely in steel and let the printed plastic do only millimetre-scale jobs:

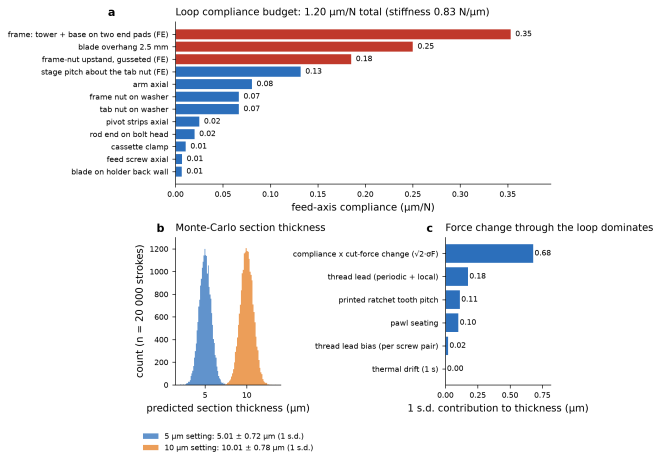
1. **Cross-strip flexure pivot**: three 0.20 mm spring-steel feeler-gauge strips carry the 125 mm arm; no sliding or rolling joint, no backlash.
2. **Differential screw from two standard threads**: M8×1.25 against M8×1.0 gives 0.25 mm per turn; a 50-tooth printed ratchet divides that into 5 µm **steps**, selectable 5–30 µm.
3. **Printed parallelogram flexure stage** carries the blade and is spring-preloaded onto the screw, so the feed has no backlash either.

A crank-rocker turns one handwheel revolution into one 54 mm stroke and one section; the arm itself advances the ratchet at the top of each stroke.

Strongest result. With every element of the force path drawn, connected and either calculated or finite-element modelled, the loop between blade edge and block face has a stiffness of $0.83\,\mathrm{N\,\mu m^{-1}}$, and a Monte-Carlo budget predicts sections of 5.0 ± 0.7 µm (1 s.d.; 90 % interval 3.8–6.2 µm) at the 5 µm setting, at a bill of materials of **US$61.97**. For context, a 154-site survey found more than two-thirds of sections from commercial microtomes outside ±10 % of the dial setting.



Revision-2 assembly: 16 printed part types (20 pieces, 1.4 L PLA) plus hardware-store steel. Frame 235 × 150 × 173 mm, one print on a 235 mm bed. Envelope 275 × 210 × 202 mm.

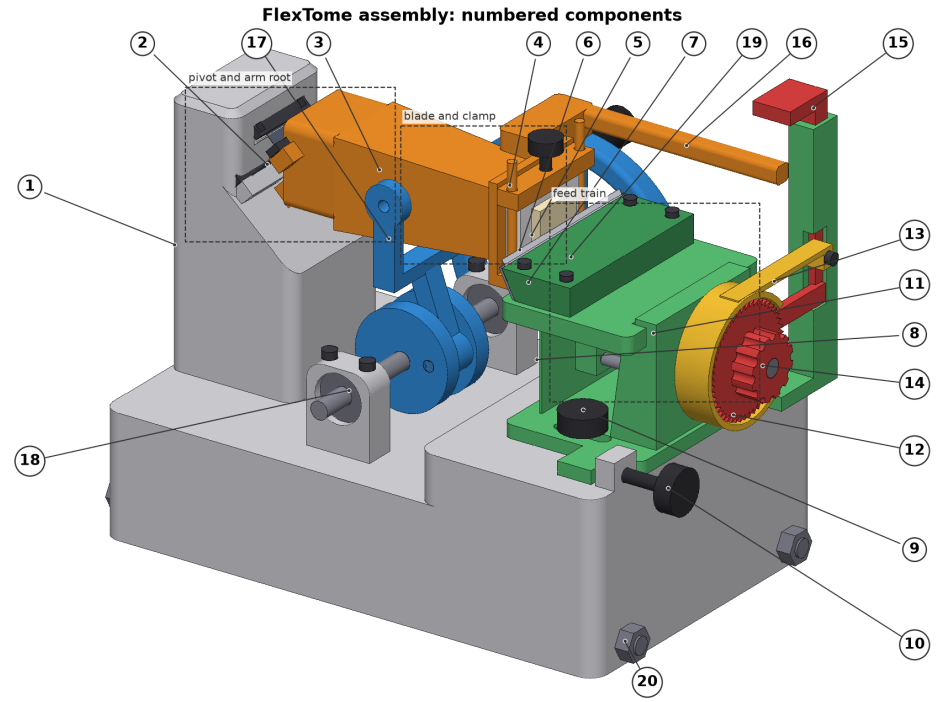


Where the compliance is (a) and what sets the thickness scatter (b, c). Stroke-to-stroke change in cutting force acting through the loop dominates; the steel feed chain contributes little.

Key numbers

Stroke / arm swing	54 mm / ±12.5°
Feed per turn / per ratchet tooth	0.25 mm / 5.0 µm
Loop compliance (stiffness)	$1.20\,\mathrm{\mu m\,N^{-1}}$ ($0.83\,\mathrm{N\,\mu m^{-1}}$)
Predicted section, 5 µm setting	5.01 ± 0.72 µm
Predicted section, 10 µm setting	10.01 ± 0.78 µm
Thread-lead bias (per machine)	within ±0.7 %
Pivot strip stress at full swing	218 MPa (fatigue SF 2.1)
First structural mode	357 Hz (cut excites <9 Hz)
Bill of materials	US$61.97

What the external review changed. An independent check of revision 1 found that its budget did not match its CAD: the feed-screw upstand was disconnected and only 6 mm thick ($7.3\,\mathrm{\mu m\,N^{-1}}$ on its own), the blade was held on two screw tips, the coupler was 0.7 mm thick, and the Monte-Carlo under-counted the force term by $\sqrt{2}$. Revision 2 gussets and connects the upstand ($0.19\,\mathrm{\mu m\,N^{-1}}$), gives the blade a full-width back wall, makes the coupler's axial path steel-on-steel, adds preload springs and washers, and re-derives every error as a stroke-to-stroke difference. Had revision 1 been built, the model says it would have cut 5 ± 4.8 µm.



1 Frame: ribbed base 235 × 150 × 55 mm, gusseted tower (one print)
2 Cross-strip flexure pivot: 3 × 0.20 mm steel strips in clamp pads
3 Rocking arm: tapered hollow box beam, 125 mm pivot to face
4 Cassette clamp: lower lip, sliding jaw, M5 thumbscrew
5 Histology cassette and paraffin block (consumable)
6 Disposable low-profile blade 80 × 8 × 0.25 mm, 27° lean
7 Blade holder (full-width back wall), 1.0 mm clamp strip, 2 × M4
8 Parallelogram flexure stage (one print), plates 1.0 × 44 mm
9 Stage clamp knobs M6 (coarse-approach slots ±10 mm)
10 Coarse-approach screw M6 in side boss
11 Differential feed screw: M8×1.0 bolt, coupler, M8×1.25 rod
12 Ratchet drum 50 T × 14 mm (5 µm per tooth) with jam nuts
13 Thickness-selector shield (1 to 6 teeth = 5 to 30 µm)
14 Feed knob
15 Pawl rod with finger, lifted by the striker bar at top of stroke
16 Striker bar on arm (feed actuation)
17 Fork connecting rod 71.2 mm and crank webs, r = 13 mm
18 Built-up crankshaft, 8 mm stubs in 2 × 608 bearings
19 Handwheel 110 mm with handle (one turn = one section)
20 M8 tie rods (base pre-compression, split-base option)

How it works in one sentence each

- *Drive*: handwheel → two-web crankshaft ($r = 13\,\text{mm}$) → fork rod → arm joint at 60 mm; $\sin\theta_{\max} = 13/60$.
- *Cut*: blade leans 27°; clearance 9.5°, rake 45.5°; the face is trimmed to the 125 mm arc once, then every section is the difference of two identical arcs.
- *Feed*: $\Delta x = (1.25 - 1.00)/50 = 5\,\mu\text{m}$ per tooth; the striker bar lifts the pawl past $\theta \approx 6°$, the pawl indexes the drum on the way down, before the face reaches the edge.
- *Why it can work in plastic*: only the force path from edge to block face sets thickness; a steady cutting force is trimmed into the block, only its stroke-to-stroke change matters.

What is still assumed, and what to measure first

- The feed-axis cutting force ($2 \pm 0.4\,\text{N}$ per stroke) is an assumption bracketed by a fracture model (0.23–0.86 N); no absolute paraffin value exists in the literature. Scatter scales almost linearly with it.
- Nut bedding on washers ($1.5 \times 10^7\,\text{N m}^{-1}$) and the PLA creep limit (15 MPa) are allowances, not measurements.
- Levers if a build cuts thick-thin: back-wall overhang 2.0 mm, base shimmed to full contact, brass inserts: together $0.90\,\mu\text{m N}^{-1}$ and 0.56 µm.
- Build-and-measure protocol (in the report): loop stiffness with a 2 N load and a dial gauge, feed per tooth over 50 teeth, then 100 consecutive 5 µm sections measured by profilometry or weighing.

Repository: CadQuery source, STEP, 35 STLs, FE and Monte-Carlo code, interference and mesh-connectivity audit, BOM, LaTeX sources.